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Group 5  
GhostDrive (Twin)

# Revision 1 System Requirements & Hazard Analysis

**Project Title:** *Human-in-the-Loop Cognitive Digital Twin of an Autonomous Scale Vehicle*

**Team:** Group 5

**Supervisor:** Dr. Istvan David

# Table of Contents

|                                                                   |    |
|-------------------------------------------------------------------|----|
| 0 Revision History.....                                           | 4  |
| 1 Purpose.....                                                    | 4  |
| 1.1 Background.....                                               | 4  |
| 2 Scope.....                                                      | 6  |
| 3 Terms and Notation .....                                        | 6  |
| 4 Four Variable Model.....                                        | 7  |
| 4.1 Monitored Variables .....                                     | 7  |
| 4.2 Controlled Variables.....                                     | 8  |
| 4.3 Constants .....                                               | 8  |
| 5 System Diagrams.....                                            | 10 |
| 5.1 Context Diagram .....                                         | 10 |
| 5.2 FSM .....                                                     | 11 |
| 5.3 Functional Decomposition Diagram .....                        | 12 |
| 5.4 Data Flow Diagram .....                                       | 13 |
| 6 Safety Requirements (Hazard Analysis) .....                     | 14 |
| 6.1 Scope and boundaries .....                                    | 14 |
| 6.2 Critical assumptions .....                                    | 14 |
| 6.3 Fault Tree Analysis (FTA) .....                               | 15 |
| 6.4 Risk mitigation & derived safety requirements .....           | 15 |
| 7 Requirements Specification .....                                | 18 |
| 7.1 Requirements Overhaul .....                                   | 18 |
| 7.2 Undesired event handling.....                                 | 19 |
| 7.3 List of requirements that are likely/unlikely to change ..... | 20 |
| 8 Roadmap (what will be implemented now vs later) .....           | 21 |
| 9 Appendix .....                                                  | 22 |
| 9.1 Appendix A: Fault Tree Analysis .....                         | 22 |
| 9.2 Appendix B: Previous Version Requirements .....               | 23 |
| 9.2.1 Performance timing requirements .....                       | 23 |
| 9.2.2 Requirements on hardware environment.....                   | 24 |
| 9.2.3 Requirements on software environment.....                   | 25 |

|                                  |    |
|----------------------------------|----|
| 9.3 Appendix C: References ..... | 26 |
|----------------------------------|----|

## 0 Revision History

| Date              | Version | Change                                                                   | Reason for change                                                                                                            |
|-------------------|---------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| November 18, 2025 | 1.0     | N/A                                                                      | N/A                                                                                                                          |
| March 18, 2026    | 1.1     | Added background section 1.1. Adapted from V&V document.                 | Provide high level description of the project.                                                                               |
| March 19, 2026    | 1.2     | Updated FSM (5.2), updated associated variable descriptions.             | To match current version used in project.                                                                                    |
| March 19, 2026    | 1.3     | Updated references section.                                              | Added reasoning for inclusion of references without direct citations.                                                        |
| March 20, 2026    | 1.4     | Updated FTA diagram (9.1) and related fault tree analysis (6.3) section. | Removed “Catastrophic Failure” section of the tree as it is not a valid hazard.                                              |
| March 20, 2026    | 1.5     | Updated risk mitigation (6.4).                                           | Changed specific values to parameterized constants.                                                                          |
| March 20, 2026    | 1.6     | Requirements overhaul (7).                                               | Overhauled requirements due to fundamental flaws (more detailed explanation in V&V document). Copied over from V&V document. |

## 1 Purpose

The digital twin system is being developed to serve as a human follower companion. Its primary purpose is to detect, track and follow a designated human by means of sensing and interpreting the surroundings. The physical twin collects and communicates information about its surroundings to the digital twin, which in turn performs computationally expensive tasks to direct the physical twin in its behaviour.

The purpose of this document is to define the hardware and software requirements for the digital twin system. It will describe the systems’ required behaviour. It will also outline the hazard analysis of the system and the safety requirements to mitigate the risk.

### 1.1 Background

The project is being developed for a HiWonder vehicle. Below is an image of the vehicle.



Figure 1: The autonomous human follower vehicle

The vehicle is equipped with two main sensors being the camera and the lidar.

The project is being developed primarily on MATLAB Simulink State flow; this portion of the project in addition to the GUI program is the digital twin.

The physical twin is the car itself. It collects and communicates information about its surroundings to the digital twin, which in turn performs computationally expensive tasks to direct the physical twin in its behavior. The physical twin is the physical human follower vehicle.

A high-level overview of the system is described in the figure below. This is for reference and not a detailed description.

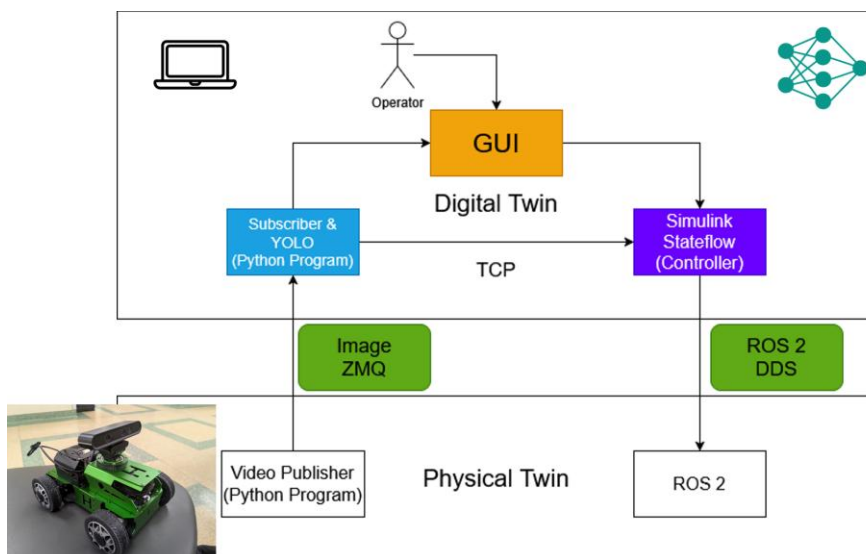


Figure 2: High-level project overview

## 2 Scope

This specification document encompasses a high-level overview of function, safety, and performance requirements for the human follower robot. It includes:

- A definition of the system's context and boundaries.
- Specifications for all monitored and controlled variables.
- A description of the required system behavior under normal and undesired conditions.
- Safety requirements derived from a formal hazard analysis.
- Performance and timing constraints.
- Identification of requirements that are stable versus those likely to change.

## 3 Terms and Notation

| Term               | Definition                                                                                                                                                                                                         |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Digital Twin       | Combination of hardware and software physically separate from the physical twin. Provides a service of performing computations which may be too expensive for the physical twin to perform in a real time setting. |
| Physical Twin      | Physical agent which interacts with the environment through various sensors and motors. Behaviour can be manually controlled or autonomous.                                                                        |
| Target person      | Designated human the system is required to follow.                                                                                                                                                                 |
| Mode               | Behaviour mode the physical twin operates in.                                                                                                                                                                      |
| Following distance | Desired distance to maintain from the target person.                                                                                                                                                               |
| Gesture            | Hand gesture which the target person shows to the physical twin to indicate desired behaviour mode.                                                                                                                |
| Obstacle           | Any object or person within the physical twin's close proximity which may prevent safe movement.                                                                                                                   |
| ROS                | Robot Operating System, key part of the software environment, the main communication backbone between the physical and the digital twin                                                                            |
| YOLO               | You Only Look Once, Object detection algorithm used to identify person and gesture perception                                                                                                                      |
| LiDAR              | Light Detection and Ranging, sensor that output point cloud that is used for obstacle detection and mapping                                                                                                        |
| FSM                | Finite State Machine, mainly used to identify the logic between changing gesture states                                                                                                                            |
| Dashboard          | The Graphical User Interface of the system(GUI) that is used to monitor the digital twin, displaying the video,                                                                                                    |

|        |                                                                                                                                                |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------|
|        | kinematics(velocity, acceleration and steer angle) and as well as the simulated surroundings                                                   |
| M_     | The prefix used for all <b>Monitored Variables</b> (e.g., M_Scan, M_RGB), which represent data coming from sensors or internal states          |
| C_     | The prefix used for all <b>Controlled Variables</b> (e.g., C_Velocity, C_Stop), which represent commands sent from the system to the hardware. |
| CONST_ | The prefix used for system-wide <b>Constants</b> (e.g., CONST_MaxV, CONST_SafeDist) that define fixed operational parameters                   |

Table 1: Used Terms and Notation

## 4 Four Variable Model

### 4.1 Monitored Variables

| Variable Name    | Description                                 | Units            | Usage                                                    |
|------------------|---------------------------------------------|------------------|----------------------------------------------------------|
| M_Scan           | 2D Lidar Scan                               | (rad, m)         | Obstacle detection and safety                            |
| M_RGB            | Raw Image from camera feed                  | Pixels           | People / gesture perception                              |
| M_Point          | Point cloud returned from depthcam/3d lidar | m                | Environment mapping / obstacle distance estimation       |
| M_Depth          | Grey scale depth image returned             | Pixels, m        | Person distance and obstacle detection                   |
| M_Battery        | Battery status                              | V<br>%           | Power monitoring and low-battery handling                |
| M_Acceleration   | Monitored acceleration from sensors         | m/s <sup>2</sup> | Comfort checks, potential safety monitoring              |
| M_DistanceApart  | Distance between vehicle and target person  | m                | Follow mode distance control                             |
| M_PersonAngle    | Relative angle to the target person         | rad              | Heading control to keep vehicle facing the target person |
| M_TargetPosition | Target person position estimate             | m, m             | Path planning / digital twin update                      |

|               |                                                                                 |                        |                                                                 |
|---------------|---------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------|
| M_YOLOPerson  | Bounding box / detection result for people from the perception module.          | [bbox, string, number] | Person detection bounding box from the vision perception module |
| M_YOLOGesture | Detected gesture label from the perception module (call, like, ok, fist, palm). | [bbox, string, number] | Detected gesture label from the vision perception module        |

Table 2: 4 Variable Model - Monitored Variables

## 4.2 Controlled Variables

| Variable Name | Description                                                           | Range                             | Units   | Usage                                                                                  |
|---------------|-----------------------------------------------------------------------|-----------------------------------|---------|----------------------------------------------------------------------------------------|
| C_Velocity    | Ros command of velocity output                                        | [-1, 1]                           | m/s     | Main speed command used in Follow / Stop behaviour                                     |
| C_Steering    | ROS Commanded steering angle / turn rate                              | $[-\frac{\pi}{4}, \frac{\pi}{4}]$ | Radian  | Heading control to keep the vehicle facing the target person or along the desired path |
| C_Stop        | Emergency stop command if obstacle within the minimum safety distance | [0, 1]                            | Boolean | Forces the vehicle into a safe stop state, overrides other commands                    |
| C_Mode        | High-level operating mode command                                     | [0, 5] (discrete)                 | N.A     | Selects overall system behaviour                                                       |
| C_Target      | Identifier of the target person                                       | [0, 1]                            | Boolean | Chooses which detected person is used for follow control and display                   |
| C_Distance    | The set following distance behind the user                            | [0.5, 2]                          | Meters  | Setpoint for Follow-mode distance control                                              |

Table 3: 4 Variable Models - Controlled Variables

## 4.3 Constants

| Variable Name | Metric | Units | Justification |
|---------------|--------|-------|---------------|
|---------------|--------|-------|---------------|



|                     |     |       |                                         |
|---------------------|-----|-------|-----------------------------------------|
| CONST_MaxV          | 1.0 | m/s   | Limit indoor speed for safety.          |
| CONST_MaxW          | 1.0 | rad/s | Avoid unstable, overly sharp turns.     |
| CONST_SafeDist      | 0.5 | m     | Minimum clearance to reduce collisions. |
| CONST_TargetTimeout | 1.0 | s     | Treat missing target as lost and stop.  |
| CONST_LidarTimeout  | 0.5 | s     | Detect LiDAR loss and fall back safely. |
| CONST_YOLOMin       | 0.8 | N.A   | Ignore low-confidence detections.       |
| CONST_Sample        | 0.5 | s     | Simulink sample time                    |

*Table 4: 4 Variable Models – Constants*

It is important to note that some variables may be added/removed based on the usefulness of them in our final design.

# 5 System Diagrams

## 5.1 Context Diagram

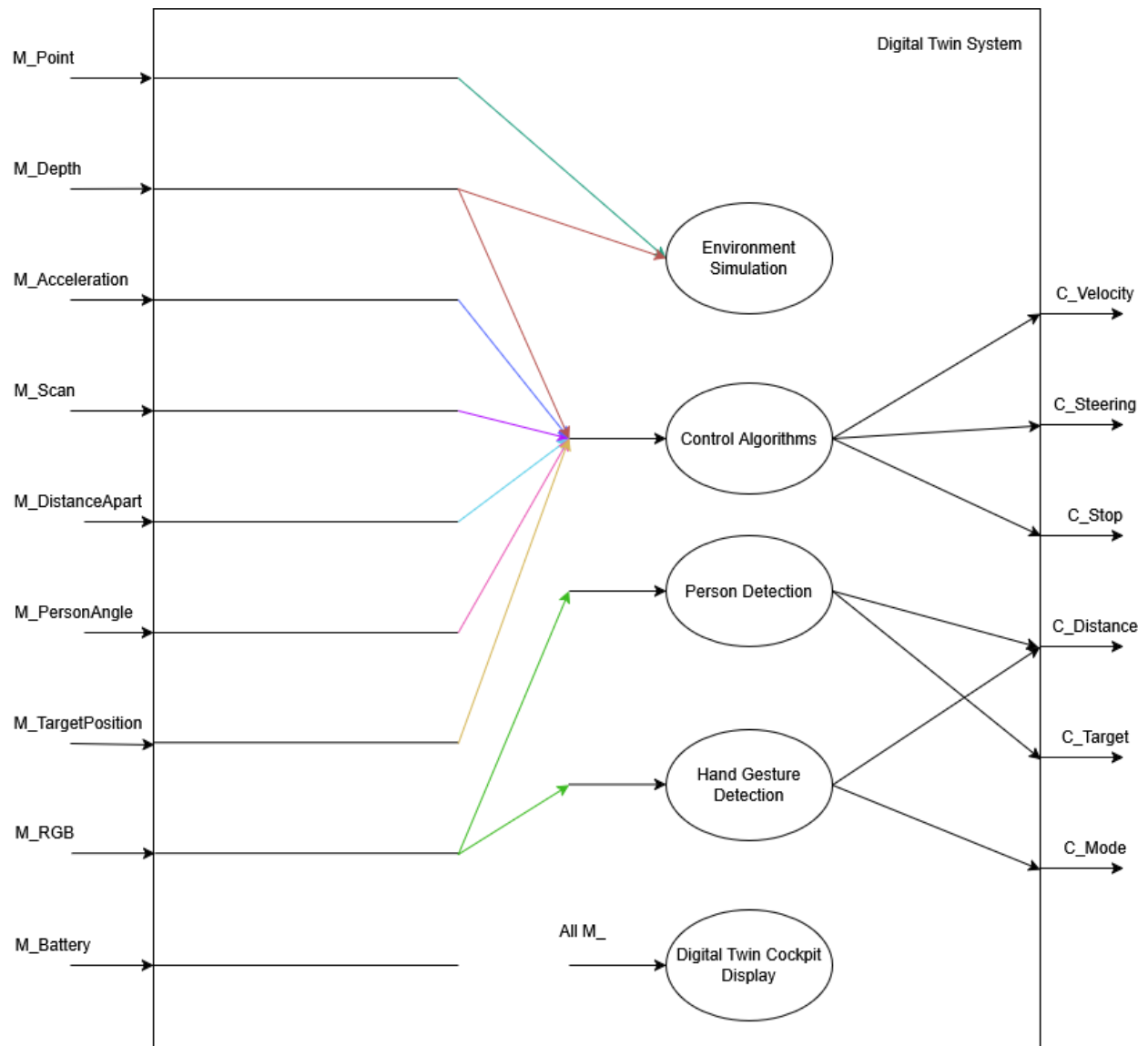


Figure 3: Context Diagram of the System

## 5.2 FSM

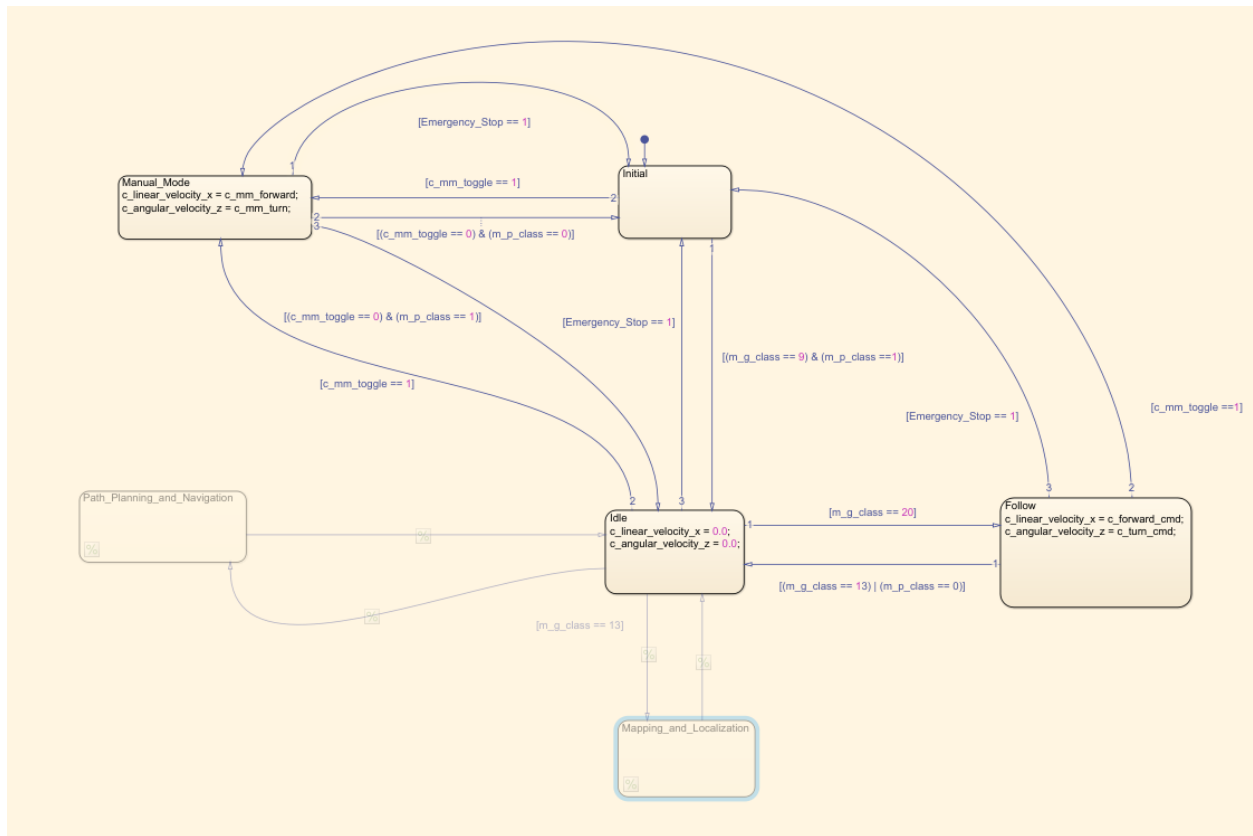


Figure 4: FSM of the System

The FSM is subject to change in the system and is expandable. The variables used (especially the specific gestures chosen) are subject to change as well.

| Variable             | Description                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| c_mm_toggle          | Toggle switch to change between manual and follower modes.                                                                                                |
| m_g_class            | The hand gesture detected by the YOLO model. m_g_class 9 is the heart gesture, m_g_class 20 is the palm gesture, m_g_class 13 is the thumbs down gesture. |
| m_p_class            | m_p_class detects the presence of a person. 1 indicates detected 0 indicates not detected.                                                                |
| c_linear_velocity_x  | Set drive forwards speed.                                                                                                                                 |
| c_angular_velocity_z | Set turn speed.                                                                                                                                           |
| Emergency_Stop       | Estop the system, 1 indicates estop pressed.                                                                                                              |

Table 5: FSM Variable Description

## 5.3 Functional Decomposition Diagram

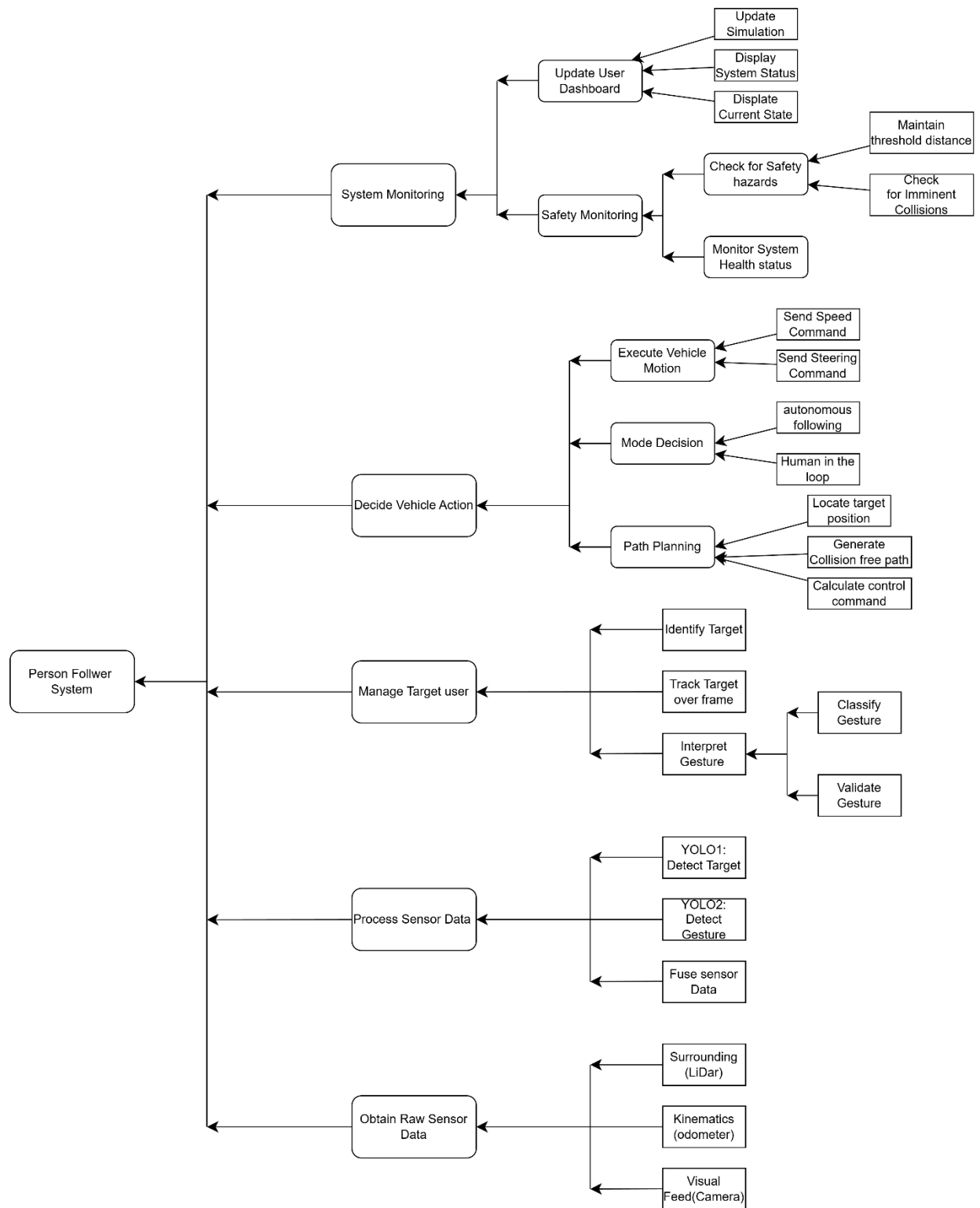


Figure 5: Functional Decomposition of the System

## 5.4 Data Flow Diagram

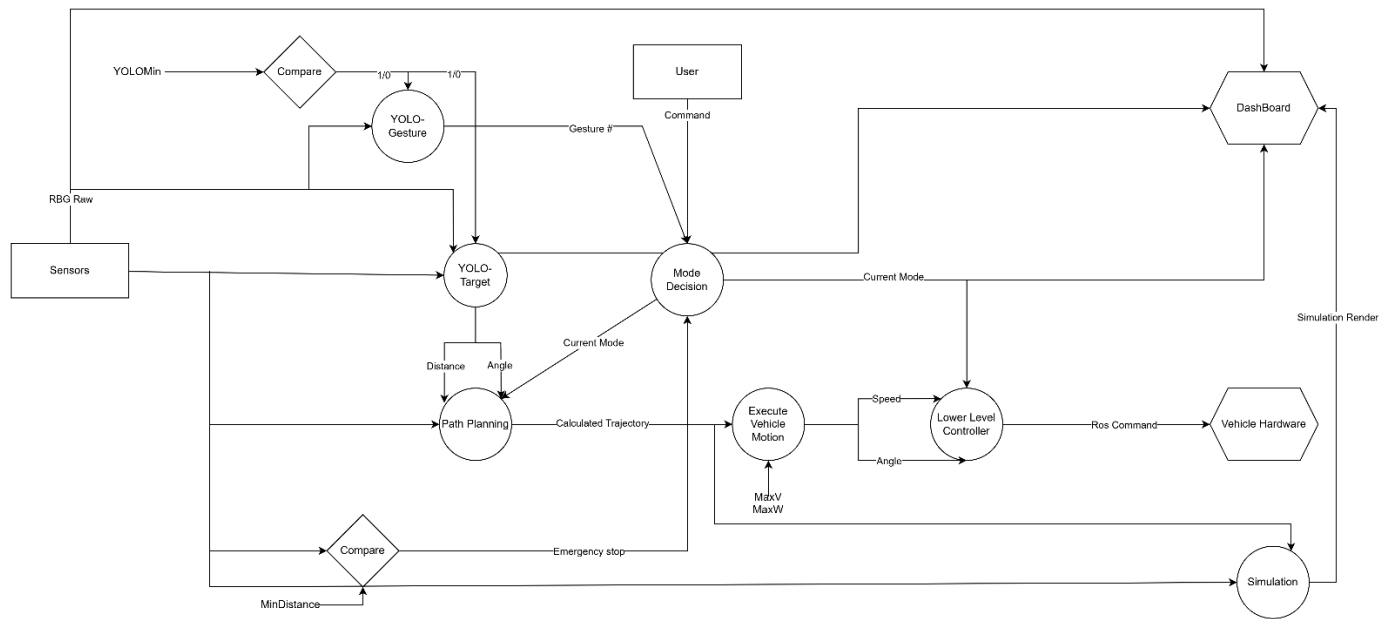


Figure 6: Data Flow Diagram of the System

The table below outlines the legend for the data flow diagram.

| Item:     | Description:                  |
|-----------|-------------------------------|
| Rectangle | Input of the overall system.  |
| Circle    | A process.                    |
| Square    | Mathematical functions.       |
| Hexagon   | Output of the overall system. |
| Arrow     | A variable.                   |

Table 6: Data Flow Diagram Legend

## 6 Safety Requirements (Hazard Analysis)

### 6.1 Scope and boundaries

The hazard analysis for this project focuses on the two major categories of risks associated with the indoor human-following autonomous vehicle: perception failure and behavioral failure. The scope includes safety risks arising from camera and LiDAR perception, YOLO detection, sensor fusion, target identification, control logic, communication, and battery or actuator malfunction.

Within this scope, the system evaluates hazards such as: failure to detect the target person, following the wrong individual, inaccurate or ambiguous gesture recognition, obstacle detection errors, incomplete environmental mapping, abnormal control commands, communication loss, and battery anomalies. Each of these failures may cause unsafe motion or unexpected system behavior.

The boundaries of the hazard analysis are defined by several explicit assumptions that limit complexity: mechanical and hardware components are assumed to function correctly; operation occurs indoors with proper lighting and a flat surface; the user behaves normally, is healthy, and provides correct gestures; wireless communication is generally stable; and no malicious interference or intentional disruption occurs. These assumptions exclude scenarios such as mechanical breakage, outdoor or harsh environments, unpredictable crowd behavior, deliberate obstruction or cyber-attacks.

### 6.2 Critical assumptions

- Pre-built hardware is working as designed
- Operation environment is indoor with flat ground surface and lighted properly
- Vehicle hardware is well maintained
- Command by target person is recognized correctly (hand gesture is recognized correctly)
- Target person is walking normally
- Human user is healthy
- Vehicle is not being attacked
- Operation is energy efficient and quiet
- Maneuverability in human crowd is sufficient
- Communication (Wi-Fi) connection is reasonably stable

## 6.3 Fault Tree Analysis (FTA)

The fault tree analysis is broken down into two top level hazards as seen in Appendix A. These were developed in accordance with our critical assumptions. They are collision, loss of tracking.

This section has been updated to remove the circular and redundant “catastrophic failure” top level hazard. All subcategories can be removed as they fall within our critical failures; except for communication failure which has been routed into loss of tracking.

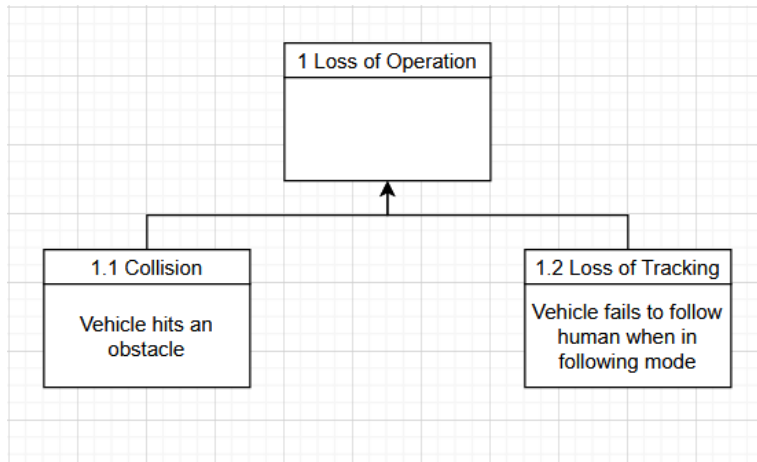


Figure 7: Top Level Hazards

Hazards are labelled in an oval shape and contain the numbering prefix letter H. As mentioned on the chart joined connections represent OR gates.

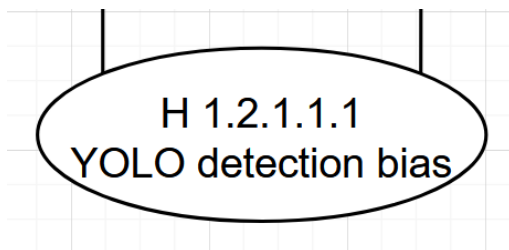


Figure 8: Example Hazard

## 6.4 Risk mitigation & derived safety requirements

This section includes the Risk Mitigation Strategies and their corresponding Derived Safety Requirements (DSRs). The risk mitigation outlines how hazards from the fault tree can be reduced or controlled to a relatively high level. The DSRs are concluded from these mitigation strategies and express the required behavior of the system using a black-box design process, which means they are externally observable, testable, and free of internal implementation details. The risk mitigations and DSRs ensure that each identified hazard is addressed with clear, verifiable safety expectations.

### 1.1 Collision

#### Risk Mitigation Strategies:

- Add redundant obstacle detection sensors.
- Validate and update the world model frequently with sensor fusion to avoid incomplete mapping.
- Use closed-loop control with feedback from encoders/IMU to detect actuator faults.
- Provide an independent emergency stop channel that stops the motor drivers.
- Avoid unsafe regions and stop if no validated path is available.
- Detect and monitor the control-update timing.

#### Derived Safety Requirements (DSR):

**DSR1:** The vehicle shall use at least two independent sensing modalities for obstacle detection. The loss of any single sensing channel does not prevent it from detecting obstacles correctly.

**DSR2:** The vehicle shall take newly observed obstacles into account in its motion decisions within MIN\_DECISION\_TIME. Otherwise, it will reduce speed or even come to an immediate stop.

**DSR3:** If the vehicle's actual speed deviates significantly from its intended motion for more than MIN\_DEVIATION\_TIME, it shall come to a controlled stop and further motion shall be stopped until the fault is cleared.

**DSR4:** The "emergency stop" (hardware button) shall cut power to the drive actuators within MIN\_ESTOP\_TIME, regardless of current operating mode or software state.

**DSR5:** The vehicle shall not enter a region which has not been verified confidently. If no such safe path exists, the vehicle shall remain stationary.

**DSR6:** The vehicle shall automatically reduce speed when the motion control updates cannot be maintained at a required frequency.

#### 1.2 Loss of Tracking

#### Risk Mitigation Strategies:

- Provide multi-modal target identification.
- Implement a "target lost" handling state instead of continuing the motion with uncertainty.
- Battery-aware follow behavior
- Constrain camera mounting and FOV to handle "camera positioned incorrectly" or "blocked by moving objects"

#### Derived Safety Requirements:



**DSR7:** The vehicle should identify the target person with a reliability sufficient to prevent following a wrong person. If the system cannot confidently identify the target person, it will not initiate.

**DSR8:** If the vehicle becomes uncertain about the target's identity or position for more than MIN\_DEVIATION\_TIME, it shall decelerate and come to a complete stop within MIN\_STOP\_TIME, entering a "Target Lost" state.

**DSR9:** If the target is not visible in the camera's intended FOV for longer than 1.5 s, the vehicle will not move toward the last predicted position until the target is redefined.

**DSR10:** The vehicle shall terminate the follow mode with a controlled stop and user notification when battery capacity becomes critically low.

**DSR11:** If the target's movement cannot be predicted reliably, the vehicle shall stop and remain stationary until reliable information is restored.

### 1.3 Battery Fire

#### Risk Mitigation Strategies:

- Contain battery-related hazards (thermal, mechanical, electrical) using physical protection.
- Detect communication loss and default to a controlled safe state rather than continuing operation.

#### Derived Safety Requirements:

**DSR12:** If the vehicle detects any abnormal battery condition, it shall immediately stop motion, disable charging/discharging operations, and switch to a safe shutdown state with a user warning.

**DSR13:** If the vehicle loses required communication with the host computer or essential control devices, it shall stop all motion and enter a pending state until communication is restored.

**DSR14:** The vehicle shall disable operation and require manual recovery whenever a fault prevents safe continued operation.

# 7 Requirements Specification

## 7.1 Requirements Overhaul

This section has been added for clarity on the changes to our requirements. The old list of requirements from V1 of this document is included in Appendix B. The following has been copied over from our V&V document where this overhaul initially took place.

A more detailed reasoning and specific changes to each requirement from V1 of this document can be found in our V&V document.

| Requirement Type  | Requirement ID | Description                                                                                                                                                                                           |
|-------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| State Behaviour   | REQ-SB-01      | The system shall not command the PT to move while in the initial state.                                                                                                                               |
|                   | REQ-SB-02      | The system shall command the PT to halt current movement while in the idle state.                                                                                                                     |
|                   | REQ-SB-03      | The system shall command the PT to follow the <b>target user at a setpoint distance defined by the user (0.5m to 2.0m)</b> while in the Follow state.                                                 |
|                   | REQ-SB-04      | The system shall not respond to motion controlling gestures unless it is inside the Idle or Follow ( <b>in future could be more</b> ) states.                                                         |
|                   | REQ-SB-05      | The system shall not command the PT to move at a linear or angular velocity above 3 m/s and 1.4 rad/s                                                                                                 |
|                   | REQ-SB-06      | The system shall command the PT to change its linear and angular velocity in accordance with user input while inside the Manual Mode state.                                                           |
|                   |                |                                                                                                                                                                                                       |
|                   | REQ-SB-07      | The system shall support distinct gesture commands to trigger specific behaviors while inside the follow state: Stop, Resume/Proceed, <b>Increase Distance (Far), and Decrease Distance (Close)</b> . |
|                   | REQ-SB-08      | The system shall accept gesture commands only from the tracked user, ignoring gestures from other persons in the field of view.                                                                       |
|                   | REQ-SB-09      | The system shall process only the most confidently detected gesture                                                                                                                                   |
|                   | REQ-SB-10      | <b>The system shall modify its path to maintain a minimum clearance of 0.2m from obstacles</b>                                                                                                        |
| State Transitions | REQ-ST-01      | The system shall start in the initial state upon initial powerup or restart.                                                                                                                          |
|                   | REQ-ST-02      | The system shall not transition out of the initial state unless a user signals to enter manual mode or displays the tracking gesture to the PT.                                                       |

|                          |            |                                                                                                                                                                                                    |
|--------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | REQ-ST-03  | The system shall not transition from the initial state to the idle state unless a user is in sight of and displays the tracking gesture to the PT.                                                 |
|                          | REQ-ST-04  | The system shall not transition to the follow state unless the system is in the idle state and the tracked user provides the follow gesture to the PT.                                             |
|                          | REQ-ST-05  | The system shall not transition from the follow state to the idle state unless the tracked user is lost or the tracked user provides the stop follow gesture.                                      |
|                          | REQ-ST-06  | The system shall transition to manual mode from any state given the user provides the manual mode input signal.                                                                                    |
|                          | REQ-ST-07  | The system shall not transition to manual mode unless the user provides the manual mode input signal.                                                                                              |
|                          | REQ-ST-08  | The system shall not exit manual mode unless the user provides the manual mode exit signal.                                                                                                        |
|                          | REQ-ST-09  | Provided the exit condition is met, the system shall transition from manual mode to the initial state if there is no current tracked person.                                                       |
|                          | REQ-ST-10  | Provided the exit condition is met, the system shall transition from manual mode to the idle state if there is a current tracked person.                                                           |
| General System Behaviour | REQ-GSB-01 | The system shall provide a <b>Manual Mode</b> where the user can directly control the PT through DT                                                                                                |
|                          | REQ-GSB-02 | The system shall allow the user to interrupt autonomous operation at any time via a hardware or software command, transitioning the vehicle to a <b>STOP</b> (initial) state within <b>200ms</b> . |
|                          | REQ-GSB-03 | The system shall enable the user to switch between "Autonomous" and "Manual" modes at any time without restarting the system.                                                                      |
| System Monitor           | REQ-SM-01  | The DT shall provide a visual interface to monitor the real-time state variables.                                                                                                                  |
|                          | REQ-SM-02  | The DT shall mirror the states and the monitored variables of the PT <b>with a maximum average latency of 500ms</b>                                                                                |

Table 7: List of requirements

The requirements that are highlighted in red are the ones that are likely to be changed as we continue the development and the testing process if the system reaches its software or hardware bottleneck.

## 7.2 Undesired event handling

| Undesired event | How we detect it | How the system responds |
|-----------------|------------------|-------------------------|
|-----------------|------------------|-------------------------|

|                             |                                                       |                                                                             |
|-----------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------|
| Lost track of target person | Target person is not detected for several cycles      | Switch to STOP mode; vehicle stops; show “Person lost” in GUI               |
| Ambiguous / false gesture   | Gesture output is low-confidence or unstable          | Ignore the gesture and keep current mode; optionally show “Gesture ignored” |
| Sensor data lost or invalid | Sensor outputs stop updating or fails basic checks    | Assume environment is unknown; switch to STOP mode until sensors recover    |
| Low battery                 | Battery level remains below a configured threshold    | Gradually reduce speed, then enter STOP mode; show Low battery              |
| Collision                   | Car deceleration detected to be above a threshold TBD | Switch to STOP mode, display “Collision Detected” in GUI                    |

Table 8: Undesired Event Handling

### 7.3 List of requirements that are likely/unlikely to change

| Requirement                             | Likelihood of Change | Possible Changes                                                                              | Rationale                                                                                                                  |
|-----------------------------------------|----------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Gesture set and gesture-to-mode mapping | High                 | Add or remove supported gestures; remap gestures to STOP / NEAR / MID / FAR modes             | User studies and supervisor feedback may show that some gestures are confusing or inconvenient to use.                     |
| Target following distances              | High                 | Adjust the desired distances for NEAR, MID, and FAR                                           | Comfortable and safe distances depend on real-world tests, walking speed, and environment constraints.                     |
| Safety distance thresholds              | High                 | Change the distance at which the system decides that something is “too close” or “safely far” | Thresholds must be tuned for different environments and sensor noise characteristics to avoid collisions and false alarms. |
| Perception and fusion parameters        | High                 | Modify confidence thresholds, timeouts, and how strictly LiDAR points                         | These parameters must balance robustness, responsiveness, and the rate of false                                            |

|  |  |                                    |                                       |
|--|--|------------------------------------|---------------------------------------|
|  |  | must align with the camera bearing | positives/negatives during operation. |
|--|--|------------------------------------|---------------------------------------|

Table 9: High Likelihood Changes

| Requirement                                             | Likelihood of Change | Possible Changes                                                                   | Rationale                                                                                                                  |
|---------------------------------------------------------|----------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| The primary behaviour of the system is person-following | Low                  | Additional behaviours may be added, but person-following remains the main use case | The project is defined around following a person; changing this would alter the core purpose of the system.                |
| ROS2 is used as the main communication backbone         | Low                  | topic names, message contents, and update rates may be adjusted                    | ROS2 is the standard middleware for this course and lab environment; replacing it is unrealistic within the project scope. |
| High-level modes are managed by an explicit FSM         | Low                  | The number or names of modes may be refined                                        | An explicit finite state machine is required for clarity, analysis, and testing of system behaviour and safety.            |

Table 10: Low Likelihood Changes

## 8 Roadmap (what will be implemented now vs later)

With the safety and security requirements established, the implementation of our roadmap is outlined as follows. Software and user-interface-related risks will be addressed during the Proof-of-Concept stage to ensure that mitigation measures are in place early, and that safety remains a central consideration throughout the project. Hazards related to the enclosure and hardware will be examined in greater detail during the prototype development phase.

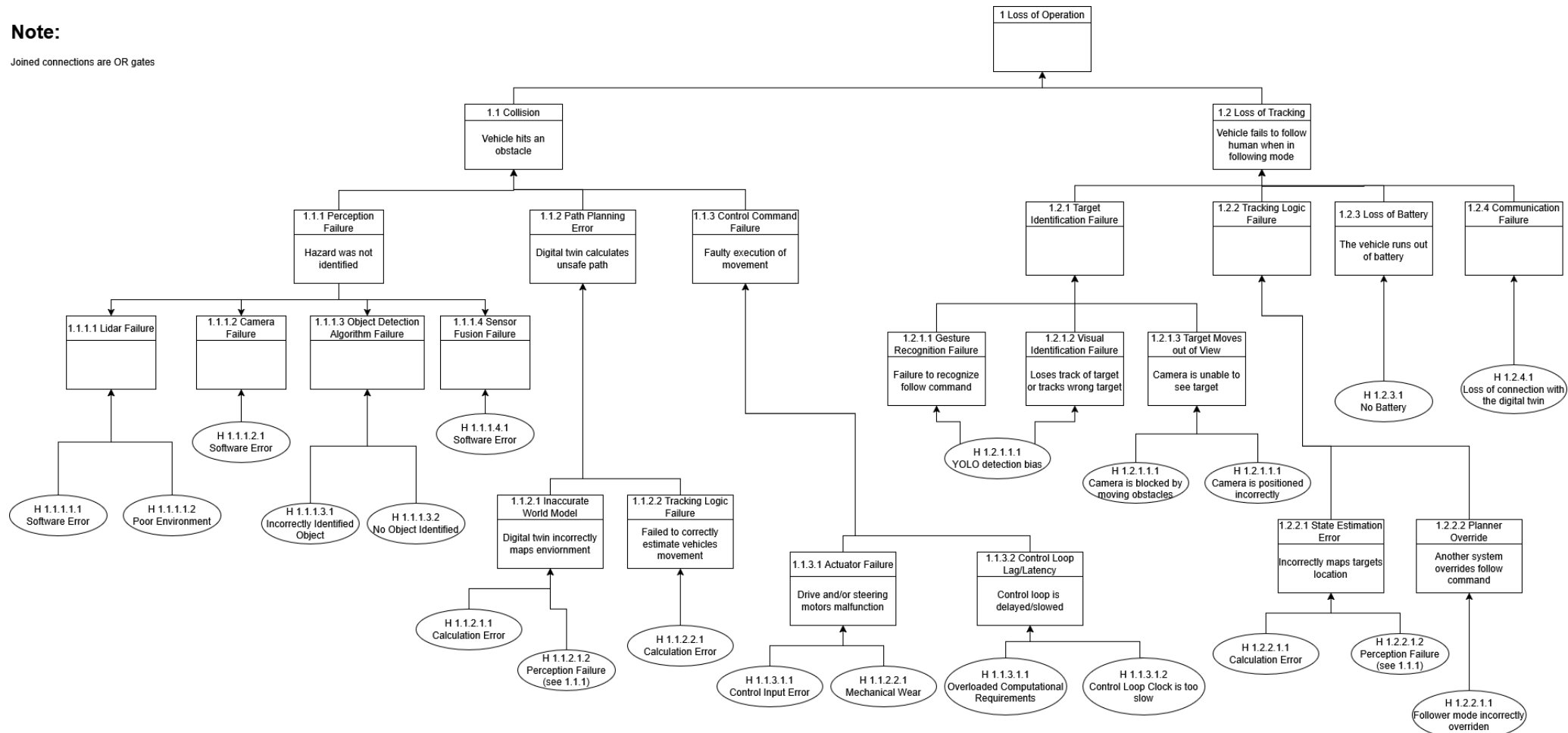
All requirements will be reassessed during the design and construction of the human tracking vehicle, taking into account project timelines and constraints. The Hazard Analysis document will serve as a continuous reference to evaluate the effectiveness of mitigation strategies for each identified risk as the design progresses. It will also be updated to incorporate any new risks or hazards that emerge as the design requirements evolve.

# 9 Appendix

## 9.1 Appendix A: Fault Tree Analysis

### Note:

Joined connections are OR gates



## 9.2 Appendix B: Previous Version Requirements

### 9.2.1 Performance timing requirements

PT-1. The core motion-control loop (reading monitored variables, computing commands, and updating controlled variables) shall run with a period of no more than 0.1 s (i.e., at least 10 Hz) during Follow mode. This ensures that changes in the target person's motion and nearby obstacles are reflected in vehicle commands quickly enough to feel smooth and safe.

PT-2. Newly observed obstacle information (for example, from M\_Scan, M\_Point, or M\_Depth) shall be considered in motion decisions within 0.1 s of being acquired. This aligns with DSR2 so that the vehicle reacts to new obstacles before entering unsafe regions.

PT-3. Under nominal conditions, the end-to-end latency from a valid change in gesture (M\_YOLOGesture) to the corresponding mode change (C\_Mode) taking effect shall not exceed 0.5 s. This keeps the system responsive to user commands and avoids confusing delays between a gesture and the observed change in behavior.

PT-4. When the target position (M\_TargetPosition or M\_DistanceApart) changes significantly, the resulting changes in C\_Velocity and C\_Steering shall occur with an end-to-end latency of no more than 0.3 s. This ensures that following behavior remains natural and prevents overshoot or oscillatory motion due to stale information.

PT-5. If the system becomes uncertain about the target identity or position, it shall detect this condition within 1.0 s and then decelerate to a complete stop within an additional 0.5 s, entering a "Target Lost" state, as required by DSR8. This limits the distance the vehicle can travel while tracking is unreliable and reduces the risk of following the wrong person.

PT-6. If any required sensor (such as LiDAR or camera) fails to update for longer than its specified timeout, the system shall transition to a safe STOP state within 0.2 s of detecting the timeout. This prevents the controller from making motion decisions based on missing or outdated sensor data.

PT-7. During Follow mode, the system shall log in key monitored and controlled variables at a rate of at least 5 Hz. This supports post-hoc analysis and verification of safety requirements and timing behavior.

PT-8. The emergency stop channel (hardware button) shall cut power to the drive actuators within 200 ms of activation,

These timing requirements are expressed independently of any specific implementation of technology. Any combination of hardware and software that can be shown by measurement, testing, or simulation to satisfy these bounds is considered acceptable.

### 9.2.2 Requirements on hardware environment

HW-1. The physical twin shall include at least two independent sensing modalities for obstacle detection (for example, a range-based sensor and a vision-based method), providing coverage within a radius of at least 2 m around the vehicle. This supports DSR1 by ensuring redundancy in obstacle perception.

HW-2. The obstacle-detection sensors shall provide range measurements with a resolution of no worse than 0.05 m and an update rate of at least 5 Hz in the primary operating region in front of the vehicle. This enables reliable detection of nearby obstacles in time to react safely.

HW-3. The system shall include at least one RGB or RGB-D camera whose field of view is sufficient to fully contain the expected target person at typical following distances (0.5–2 m), with a nominal frame rate of at least 10 fps under indoor lighting. This provides adequate visual coverage and temporal resolution for person and gesture perception.

HW-4. The drive system shall support independent control of forward velocity and steering (or equivalently, differential wheel speeds), with command resolution fine enough that each command step can change velocity by no more than 0.05 m/s and steering angle by no more than 0.05 rad. This allows the controller to implement smooth, incremental adjustments to heading and speed.

HW-5. The vehicle shall be equipped with wheel encoders and/or an IMU capable of estimating linear velocity with an error of no more than 0.05 m/s over one-second intervals. This supports closed-loop feedback and detection of actuator faults in line with DSR3.

HW-6. A hardware emergency stop mechanism shall be physically accessible to the operator while walking near the vehicle and shall interrupt power to the drive actuators independently of the main control computer. This provides a last-resort safety mechanism that does not rely on software.



HW-7. The on-board power system (battery and power distribution) shall supply stable voltage and current sufficient for continuous operation of all sensors, computer hardware, and actuators for at least the planned demonstration duration. This reduces the risk of unexpected brownouts or sudden shutdowns that could lead to unsafe behavior.

HW-8. The communication link between the physical and digital twin (for example Wi-Fi or an equivalent technology) shall support a sustained data rate sufficient to carry all required sensor streams and control commands while respecting the timing requirements defined in Section 7.1. This ensures that remote computation does not violate latency and update-rate constraints.

HW-9. All safety-critical connections, such as the emergency stop circuitry, motor driver enable lines, and battery monitoring connections, shall be electrically robust and shall be designed to tolerate single-point faults such as loose connectors without silently disabling safety functions. This reduces the risk that a simple hardware fault prevents safe operation.

These hardware requirements do not mandate any specific sensor model, computing platform, or motor type. Any hardware platform that provides the capabilities described above is considered acceptable.

### 9.2.3 Requirements on software environment

SW-1. The runtime environment shall provide a publish–subscribe communication middleware (for example ROS 2) that supports reliable message passing for all monitored and controlled variables between the physical twin and the digital twin. This enables perception, control, and GUI components to exchange data in a modular and observable way.

SW-2. The operating system shall provide a system clock with a resolution of at most 1 ms, and the software shall timestamp sensor messages and control commands using this clock. This allows the system to measure and enforce the timing requirements described in Section 7.1.

SW-3. The software environment shall support deep-learning inference for person and gesture detection on the incoming camera stream at a sustained effective rate of at least 5 Hz under nominal conditions. This ensures that M\_YOLOPerson and M\_YOLOGesture are updated frequently enough for responsive following and mode switching.

SW-4. The middleware shall support quality-of-service (QoS) configuration or equivalent mechanisms that allow safety-critical topics (such as obstacle data, emergency stop, and velocity commands) to be prioritized over non-critical topics (such as the GUI video feed) when computing or communication resources are limited. This ensures that degraded performance affects non-critical functions first, thereby preserving safety.

SW-5. The software stack shall provide a configurable logging and replay facility capable of recording key monitored and controlled variables at the rates specified in PT-8. This enables debugging, verification of safety behavior, and offline analysis of incidents.

SW-6. The GUI (dashboard) shall run as a separate process or node that subscribes to system state topics without blocking the real-time control and perception loops. This prevents visualization tasks from interfering with safety-critical timing.

SW-7. The environment shall provide a simulation or test harness, such as playback of logged data or a simple virtual environment, that allows the control logic and safety requirements to be exercised without requiring a physical vehicle. This facilitates early validation of requirements and regression testing as the design evolves.

## 9.3 Appendix C: References

The references mentioned are resources we have used in developing the System Requirements & FTA. Specific statements/information has not been directly taken from these sources.

[1] [S. Ramdhan, I. David et al], "[Engineering Automotive Digital Twins on Standardized Architectures: A Case Study]," *arXiv preprint arXiv:2508.18662*, Aug. 2025. [Online]. Available: <https://arxiv.org/abs/2508.18662>. [Accessed: Nov. 18, 2025].

[2] ISO 23247, "Digital Twin Manufacturing Framework," *ap238.org*, [Online]. Available: <https://www.ap238.org/iso23247/>. [Accessed: Nov. 18, 2025].

[3] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018. [Online]. Available: <http://incompleteideas.net/book/RLbook2020.pdf>.

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